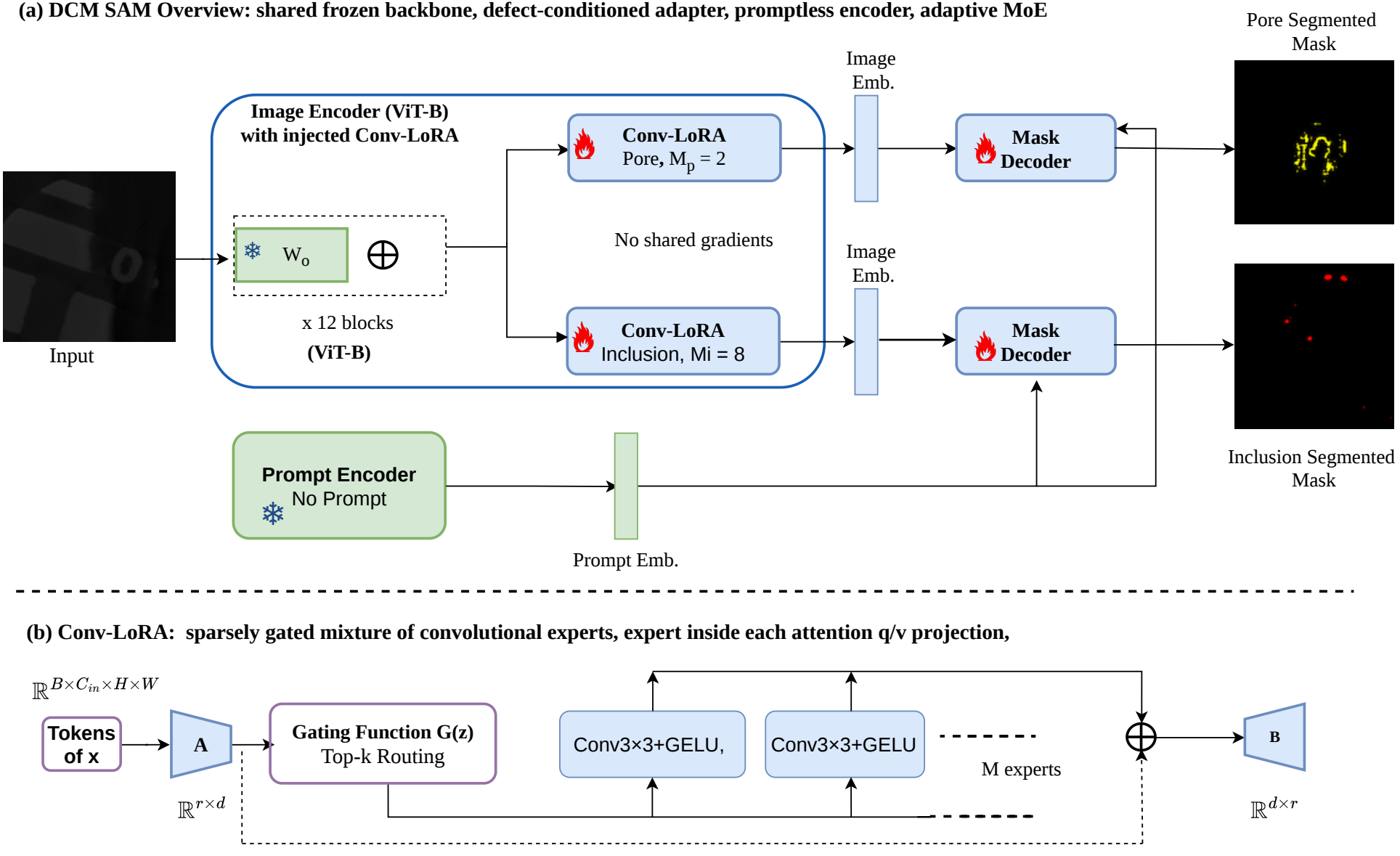

(a) DCM SAM Overview: shared frozen backbone, defect-conditioned adapter, promptless encoder, adaptive MoE
Pore Segmented Mask
Image Emb.
Image Encoder (ViT-B) with injected Conv-LoRA
Conv-LoRA Pore, M_p = 2
Mask Decoder
Input
W_o
No shared gradients
x 12 blocks
(ViT-B)
Image Emb.
Conv-LoRA Inclusion, Mi = 8
Mask Decoder
Prompt Encoder
No Prompt
Prompt Emb.
Inclusion Segmented Mask
(b) Conv-LoRA: sparsely gated mixture of convolutional experts, expert inside each attention q/v projection,
$\mathbb{R}^{B\times C_{in}\times H\times W}$
Tokens of x
A
$\mathbb{R}^{r\times d}$
Gating Function G(z)
Top-k Routing
Conv3×3+GELU,
Conv3×3+GELU
M experts
B
$\mathbb{R}^{d\times r}$